**Assignment 1 (Due: Nov. 8, 2026)**
(Send your solutions to our TA, Linfei Li, cslinfeili@tongji.edu.cn)

1. (**Math**) A binary classification test set contains 100 samples: 20 positive and 80 negative. The prediction results of classifiers A and B are shown below.

| Classifier | TP | FP | FN | TN |
|---|---|---|---|---|
| A | 8 | 2 | 12 | 78 |
| B | 16 | 12 | 4 | 68 |

Here, TP, FP, FN, and TN denote true positives, false positives, false negatives, and true negatives, respectively. Calculate the accuracy, precision, recall, and $F_1$ score for each classifier. Does selecting a classifier based on accuracy lead to the same conclusion as selecting one based on the $F_1$ score?
Let $P$ and $R$ denote precision and recall, respectively, with $P, R > 0$. The $F_1$ score is defined as $F_1 = \dfrac{2 \times P \times R}{P + R}$

2. (**Math**) In our lecture, we mentioned that for logistic regression, the cost function is,

$$J(\boldsymbol{\theta}) = -\sum_{i=1}^{m} y_i \log\left(h_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\right) + (1 - y_i)\log\left(1 - h_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\right)$$

Please verify that the gradient of this cost function is

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \sum_{i=1}^{m} \boldsymbol{x}_i \left(h_{\boldsymbol{\theta}}(\boldsymbol{x}_i) - y_i\right)$$

3. (**Math**) Please prove the Proposition 20 mentioned in our lecture.

**Proposition 20**:




Suppose that $\mathbf{x}^*$ and $(\alpha^*, \beta^*)$ are primal feasible and dual feasible respectively. If the duality gap associated with them is 0, i.e.,

$$f_0(\mathbf{x}^*) - g(\alpha^*, \beta^*) = 0$$

Then, $\mathbf{x}^*$ and $(\alpha^*, \beta^*)$ should be primal optimal and dual optimal, respectively, and the primal problem has strong duality

4. (**Programming**) Why Does Classifying Concentric Circles Require a Nonlinear Decision Boundary?
Use the following code to generate a dataset in which the two classes form concentric circles:

```
from sklearn.datasets import make_circles
from sklearn.model_selection import train_test_split
```

```
X, y = make_circles(
        n_samples=600,
        factor=0.4,
        noise=0.10,
        random_state=42
)

X_train, X_test, y_train, y_test = train_test_split(
        X, y,
        test_size=0.3,
        stratify=y,
        random_state=42
)
```

(1) Visualizing the data

Create a scatter plot of the samples, using different colors for the two classes. Based on the plot, explain whether a straight line can effectively separate the two classes.

(2) Training and comparing classifiers

Train the following three classifiers and report the training and test accuracy of each. You may use existing model-training APIs directly. For Model B, transform each two-dimensional sample into a single scalar feature. Apply the same transformation to both the training and test sets.

| **Model** | **Input features** | **Parameter settings** |
|---|---|---|
| A: Linear SVM | Original features ($x_1$, $x_2$) | SVC(kernel="linear", C=1) |
| B: Linear SVM | New feature $z = x_1^2 + x_2^2$ | SVC(kernel="linear", C=1) |
| C: RBF-kernel SVM | Original features ($x_1$, $x_2$) | SVC(kernel="rbf", C=1, gamma=1) |

(3) Interpreting the results

Based on your experimental results, answer the following questions:

- What is the geometric meaning of the feature $z = x_1^2 + x_2^2$? Why does it help distinguish the two classes?
- Suppose Model B has a decision threshold of $z = t$ in the new feature space, where $t > 0$. What shape does this decision boundary take in the original two-dimensional space?
- Both Model B and Model C can handle nonlinear classification problems. How do they differ in the way they use features?